LatticeSense IQ-40 Environmental Sensor

Continuous indoor air-quality monitoring for commercial buildings, laboratories, and light industry.

Overview

The IQ-40 is Kestrel's fourth-generation ambient sensor node. It samples particulate matter, CO2, VOC, temperature, and relative humidity on a single board and streams calibrated readings over LoRaWAN every 30 seconds. A ten-year field-replaceable battery and tool-free mounting bring installed cost below half that of wired alternatives.

Each unit ships factory-calibrated against NIST-traceable references and re-calibrates optically in the field using Kestrel's drift-model firmware, holding PM2.5 accuracy within ±8% for the service life without bench recalibration.

Signal Path

Sensing, aggregation, and uplink are handled on-device; no local gateway software is required beyond a standard LoRaWAN network server.

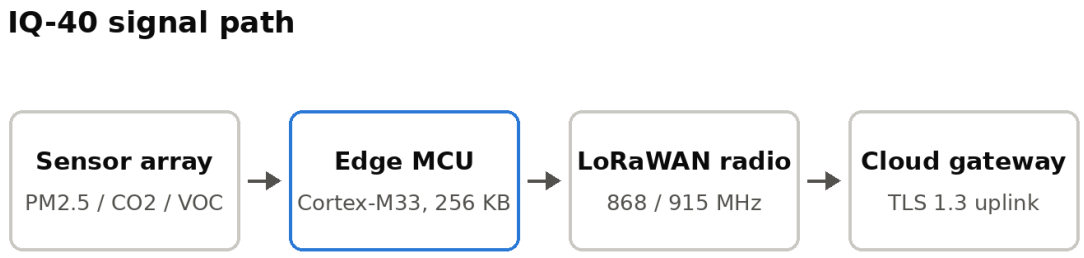


Figure 1 — IQ-40 acquisition and uplink chain.

Key Capabilities

- Five-parameter sensing (PM2.5, CO2, VOC index, temperature, RH) on one node, one radio licence.
- 30-second reporting interval at a 10-year battery life (2 AA lithium primary cells, field-replaceable).
- Optical drift compensation — no scheduled bench recalibration over the service life.

- Native LoRaWAN 1.0.4 with OTAA join; EU868 and US915 SKUs.
- REST and MQTT delivery from Kestrel Cloud with per-site data residency options.

Telemetry is retrievable per device, for example `GET /v1/telemetry?device=iq40-000342&window=24h` against the site tenant endpoint.

Model Comparison

The table compares the IQ-40 with the prior-generation IQ-25 and the closest competitive node in its price band, the Boreal ES-3.

Specification	IQ-40	IQ-25 (prior gen)	Boreal ES-3
Sensed parameters	PM2.5, CO2, VOC, T, RH	CO2, T, RH	PM2.5, CO2, T, RH
Reporting interval	30 s	120 s	60 s
Battery life	10 years	5 years	3 years
Radio	LoRaWAN 1.0.4	LoRaWAN 1.0.2	Wi-Fi 4
Field recalibration	Not required (optical drift model)	Every 24 months	Every 12 months
Ingress rating	IP54	IP42	IP40
List price (100+ units)	USD 148	USD 129	USD 165

Availability

EU868 units ship from stock; US915 units ship from week 36. Evaluation kits (three nodes, one indoor gateway, 90-day cloud tenancy) are available through certified integration partners. Volume pricing and the integration guide are provided under NDA by regional sales.

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